Decision Trees and Random Forests

Deep Learning for Scientific Discovery, Lecture 9

Marcin Płodzień

Atomic Optics Department
Institute of Theoretical Physics
Jagiellonian University in Kraków

2026/2027



Learning route

We learn to score a question, grow a tree and trace a prediction. We then predict numerical targets, choose complexity on validation observations and compare a forest and a network.

- Fractions and averages are enough for the numerical examples.
- The foundations notebook uses supplied galaxy measurements and needs no image download.
- Detailed tree derivations and image measurements are available in the notes. Further slides retain selected tree formulas.

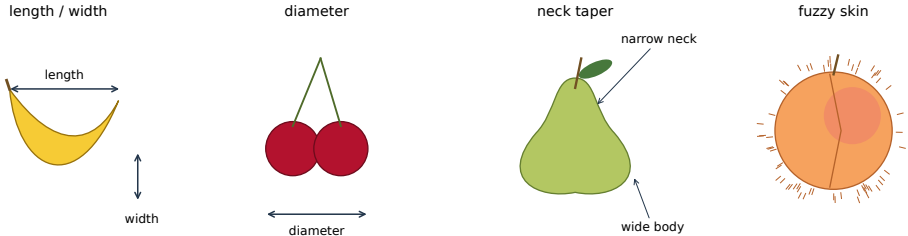
Pause at the practice frames and calculate before reading the feedback.

Where we are

In Lectures 3 to 7 every model was a neural network. Networks learned labels from images (Lecture 4), structure without labels (Lecture 5), new samples (Lecture 6) and forecasts of time series (Lecture 7).

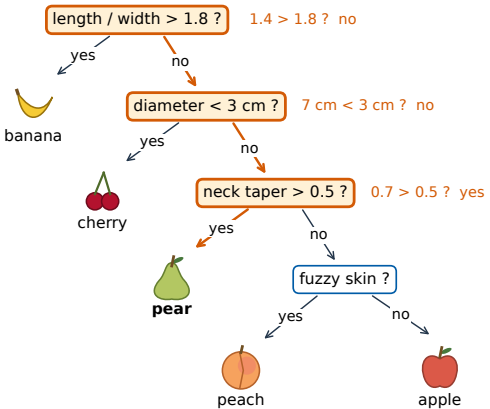
Today the data are tables of measured quantities, and the models are built from questions about those quantities. We ask what such models learn, how they can be read, and when they are enough without a network.

What the questions measure



Bananas, cherries, pears, peaches and apples, each described by four measurements: the ratio of length to width, the diameter, the neck taper (how strongly the body narrows towards the stalk) and whether the skin is fuzzy.

A key of if-else questions



A greengrocer's key as a program:

```
if length/width > 1.8:    banana
elif diameter < 3 cm:     cherry
elif neck taper > 0.5:    pear
elif fuzzy skin:          peach
else:                     apple
```

A pear with length/width 1.4, diameter 7 cm and neck taper 0.7 answers no, no, yes and reaches the leaf pear. The skin is never asked about.

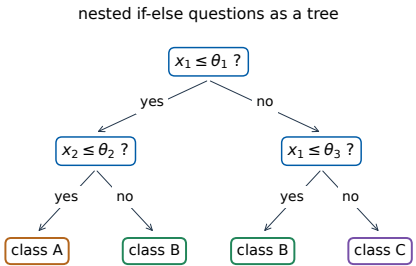
Decision trees

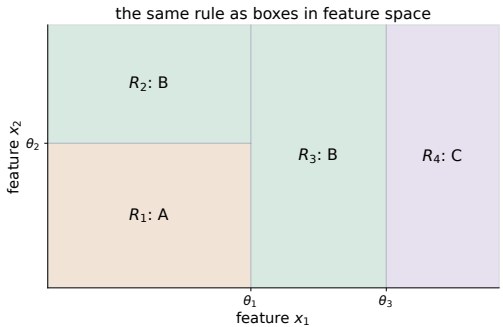
Each internal node asks one question about one measurement, each answer selects the next question, and each leaf names a class. Models of this form are called **decision trees**. A person wrote this key, knowing which measurements matter. A tree trained on labelled examples learns the key itself: which question to ask first, about which measurement, at which threshold, what to ask next in each branch, when to stop, and what to predict in each leaf.

A neural network computes weighted sums of all inputs. A tree asks one question at a time, and the learned questions can be read and checked.

L. Breiman, J. H. Friedman, R. A. Olshen, C. J. Stone, *Classification and Regression Trees*, Wadsworth (1984)

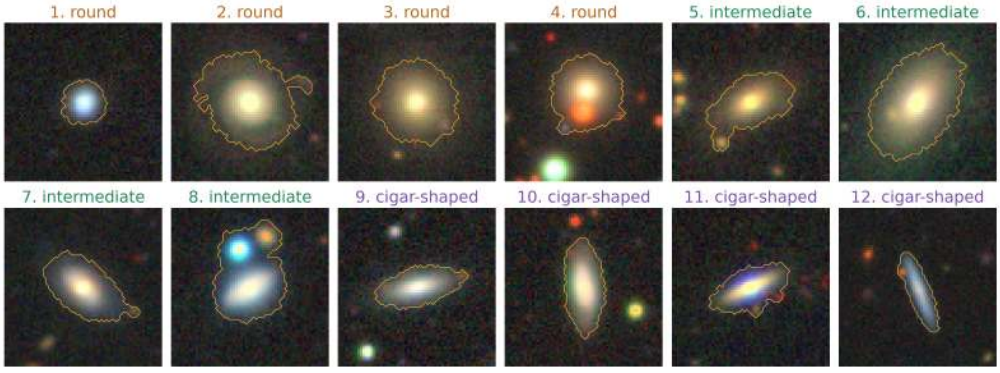
A tree divides feature space into boxes





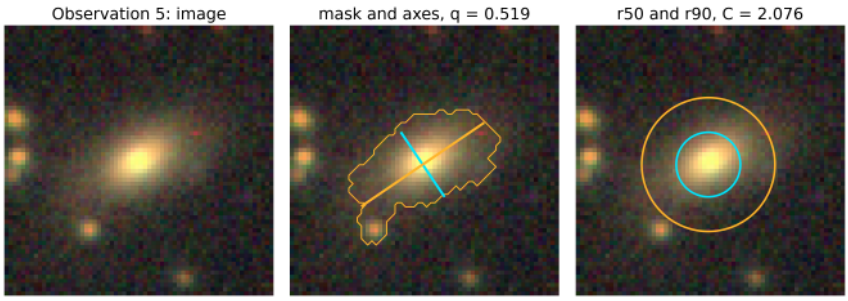
Each leaf owns a box R_ℓ and predicts a constant c_ℓ there, $f(\mathbf{x}) = \sum_\ell c_\ell \, \mathbf{1}\{\mathbf{x} \in R_\ell\}$.
Learning a tree means choosing the boxes and the constants.

Twelve galaxies of the fitting set



The first four galaxies of each class in the fitting set, the galaxies used for training, numbered. Orange outlines are the measured masks. The tree is trained on all 3407 fitting galaxies.

From an image to a feature row



The supplied column q measures roundness. Larger values mean a rounder image. The supplied column C compares radii enclosing 90% and 50% of the masked display brightness. Both are dimensionless. The notes derive these measurements in an appendix.

These columns are computed from the image without its label. The catalogue supplies only the label.

Fitting, validation and test galaxies

Role	Round	Intermediate	Cigar-shaped
Fitting: 3,407	1797	1390	220
Validation: 853	450	348	55
Test: 746	398	289	59

The validation galaxies are drawn from the non-test galaxies by stratified sampling, separately within each class, so that both parts keep the class proportions. The fitting rows determine splits and leaf predictions. Validation galaxies choose settings, and test galaxies describe the chosen model.

One fitting mask failed. Its missing features are replaced by the fitting medians, a replacement of missing values called imputation, and a failure flag records it. No validation or test values enter the imputation.

Fixed course split. The test galaxies were already used in Lecture 6.

The structure of a decision tree

The **root** holds all fitting observations. An internal **node** asks a question $x_j \leq \theta$. Its two **branches** lead to children, yes on the left and no on the right.

A **leaf** stores a prediction. The **depth** counts comparisons from the root, and our first tree permits at most two.

We have not chosen any thresholds yet. Training compares candidate questions on the fitting rows that reach each node. Prediction sends a new input along the stored questions to one leaf, and its target is never used.

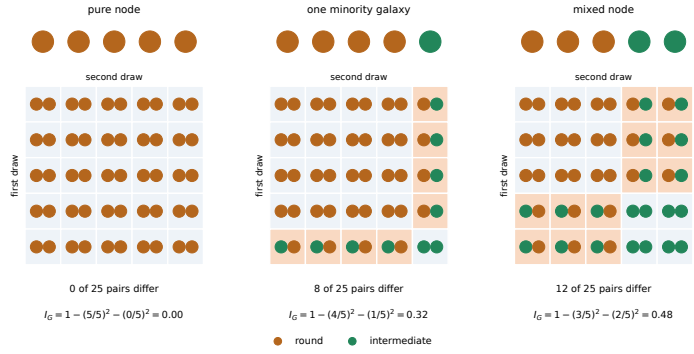
Class proportions inside a node

A node t holds n_t fitting observations, and n_{tk} of them belong to class $k = 1, \ldots, K$. The class proportions are

$$p_{tk} = \frac{n_{tk}}{n_t}, \qquad \sum_{k=1}^{K} p_{tk} = 1.$$

The galaxy root has counts $(1797, 1390, 220)$ in the order round, intermediate, cigar-shaped. Its proportions are these counts divided by 3407. A pure node contains one class, and this root is mixed. A leaf stores its proportions as class probabilities and predicts its most frequent class.

All pairs of five galaxies



Three nodes of five galaxies (brown round, green intermediate). Drawing two galaxies with replacement gives 25 equally likely ordered pairs, one per cell (row first draw, column second). Orange cells hold pairs with different labels, 0, 8 and 12 of 25. This fraction is the Gini impurity, $1 - \sum_k p_{tk}^2 = 0$, 0.32 and 0.48.

The Gini impurity

For two independent labels Y, Y' drawn with the class proportions p_{tk} of node t,

$$I_G(t) = P(Y \neq Y') = \sum_{k=1}^{K} p_{tk}(1 - p_{tk}) = 1 - \sum_{k=1}^{K} p_{tk}^2.$$

After drawing class k, the second label differs with probability $1 - p_{tk}$. Both labels equal k with probability p_{tk}^2.

A pure node has $I_G = 0$. Equal proportions of K classes give the largest value, $1 - 1/K$.

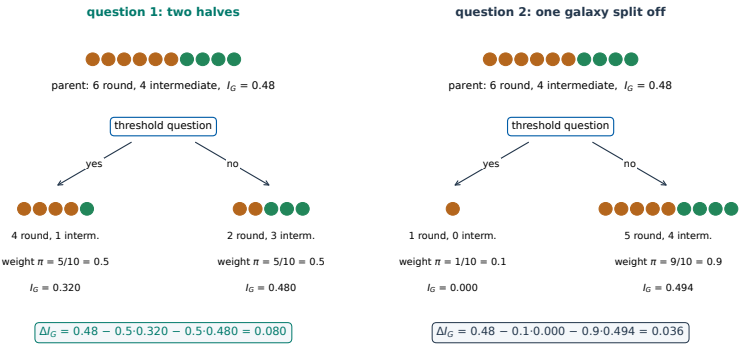
Practice. A question for ten labels

A node has six A and four B labels. Question U gives children $(4, 1)$ and $(2, 3)$. Question V gives $(1, 0)$ and $(5, 4)$.

1. Calculate the Gini impurity of the parent and each child.
2. Weight each child by its share of the parent and choose a question.
3. State which observations and labels enter this calculation.

The next frame supplies the weighted comparison.

Child sizes determine their influence



A node of six round and four intermediate galaxies, $I_G = 0.48$. Each child impurity is weighted by its share $\pi = n_{\text{child}}/n$. Splitting off one galaxy gives a pure child of weight 0.1, while the child of weight 0.9 keeps nearly the parent impurity, so its decrease is 0.036 against 0.080 for two different halves.

The split objective

A question $x_j \le \theta$ sends the n observations of node t to a left child L (yes) and a right child R (no). With the child fractions $\pi_L = n_L/n$ and $\pi_R = n_R/n$, the decrease of an impurity I is

$$\Delta I = I(t) - \pi_L\, I(L) - \pi_R\, I(R).$$

Training selects the feature j and threshold θ with the largest ΔI.

Our single-tree classifiers use Gini impurity, for which the decrease takes the form

$$\Delta I_G = \pi_L \pi_R \, \|\boldsymbol{p}_L - \boldsymbol{p}_R\|^2.$$

The gain rewards different class proportions in two substantial groups. The notes derive this identity after the trained-tree example. The misclassification error $1 - \max_k p_{tk}$ and the entropy are alternative impurities, compared in the appendix.

Recursive tree growth

1. Compare all allowed questions using the observations in the current node.
2. Choose the question with the largest impurity decrease and pass the observations to the two children.
3. Repeat inside each child until a stopping rule applies, for example a maximum depth or a minimum number of observations per leaf.
4. Store the class proportions, or the mean response, at each leaf.

The ideal tree would minimise $\sum_\ell n_\ell I(R_\ell)$ over all divisions into boxes, a search far too large. **Recursive binary splitting** is greedy instead. Each question is the best one for its own node and is never revisited, so the complete tree need not be the best possible tree.

G. James, D. Witten, T. Hastie, R. Tibshirani, *An Introduction to Statistical Learning*, Springer (2013), Ch. 8, Sec. 8.1.1

Impurity of the galaxy root

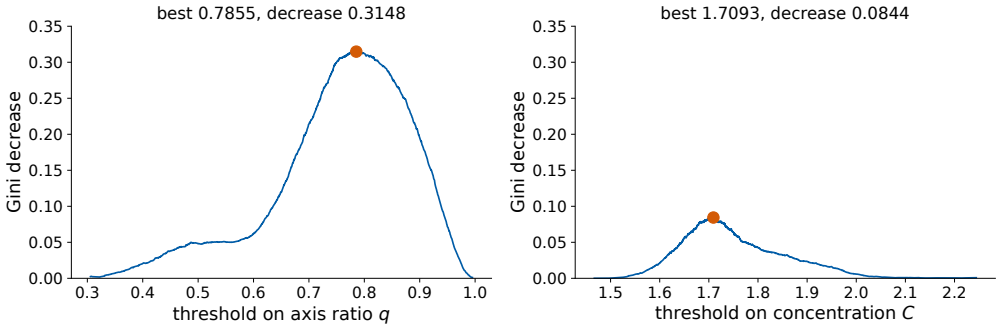
At the galaxy root, the fitting counts are $(1797, 1390, 220)$ and $n = 3407$.

$$I_G(t) = 1 - \left(\frac{1797}{3407}\right)^2 - \left(\frac{1390}{3407}\right)^2 - \left(\frac{220}{3407}\right)^2 = 0.551183.$$

For each feature, sort its values and try midpoints between neighbouring distinct values. Require at least ten observations in each child.

For every candidate, count the labels on both sides, compute their impurities and compare the weighted decrease.

Search over features and thresholds



The best q question gains 0.314761 and the best C question 0.084387. The failure flag cannot isolate a valid leaf of ten rows. The root therefore asks about q.

The first learned question

The winning question is $q \le 0.785473$. Count vectors have order round, intermediate, cigar-shaped.

Answer	Class counts	Size	Gini
Yes	$(203, 1332, 219)$	1754	0.394317
No	$(1594, 58, 1)$	1653	0.068880

$$\Delta I_G = 0.551183 - \frac{1754}{3407}(0.394317) - \frac{1653}{3407}(0.068880) = 0.314761.$$

The children are still mixed. The question has formed groups for the next step, and the classification continues inside them.

The search inside each child

The left child contains only the 1,754 galaxies that answered yes. Its best question is $q \leq 0.515985$:

$$(203, 1332, 219) \longrightarrow (37, 77, 193) \quad \text{and} \quad (166, 1255, 26).$$

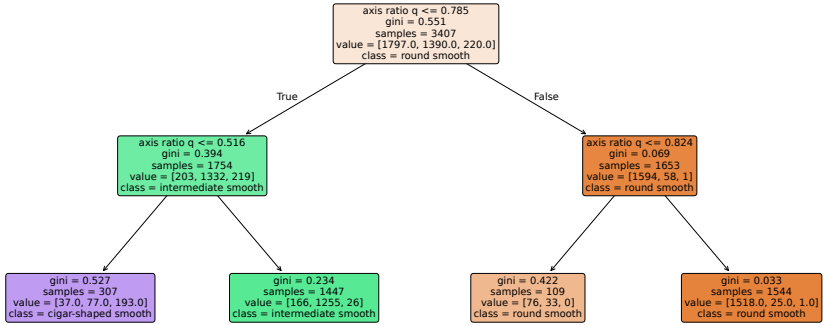
The majority classes become cigar-shaped and intermediate.

The right child searches its own 1,653 rows and chooses $q \leq 0.823698$:

$$(1594, 58, 1) \longrightarrow (76, 33, 0) \quad \text{and} \quad (1518, 25, 1).$$

Both leaves predict round, but their class probabilities differ. All four leaves reach depth two, so the prescribed stopping rule ends training.

The trained depth-two tree



Every chosen split uses q, although C was eligible. Labels overlap, so the leaves remain mixed. Thresholds are rounded in the diagram.

A prediction follows one path

The first validation galaxy, row 4234, has $q = 0.908681$. Its catalogue label is put aside until the prediction is made.

1. The root asks $q \leq 0.785473$. The answer is no, and the galaxy follows the right branch.
2. The right child asks $q \leq 0.823698$. The answer is again no, and the galaxy reaches the rightmost leaf.

That leaf contains $(1518, 25, 1)$ fitting galaxies, so

$$\widehat{\boldsymbol{p}}(x) = \frac{(1518, 25, 1)}{1544} \simeq (0.98316, 0.01619, 0.00065).$$

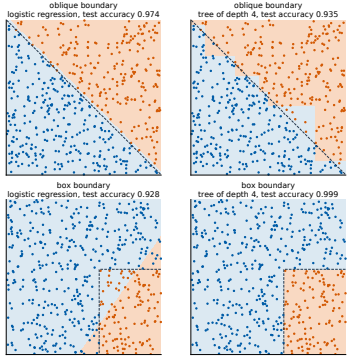
The tree predicts round smooth, which is also the catalogue label. The leaf fractions are frequencies among 1544 fitting galaxies, and their calibration has to be checked on new data.

Practice. Probabilities and decisions

The rounder branch has counts $(1594, 58, 1)$ and splits into $(76, 33, 0)$ and $(1518, 25, 1)$.

1. Predict a hard class in each leaf and in the parent.
2. State what changes if this last question is removed.
3. Name a score that can register the probability change.

Boxes or straight lines



Test accuracy on 5,000 points. An oblique boundary favours logistic regression (0.974 against 0.935 for the tree staircase), a box-shaped class favours the tree (0.999 against 0.928). Trees build boundaries from axis-parallel pieces.

After G. James, D. Witten, T. Hastie, R. Tibshirani, *An Introduction to Statistical Learning*, Springer (2013), Ch. 8, Fig. 8.7

Scores on held-out galaxies

The balanced accuracy averages the recall of the K classes,

$$\mathrm{BA} = \frac{1}{K} \sum_{k=1}^{K} \frac{T_k}{N_k},$$

with N_k evaluated observations of class k, of which T_k are predicted correctly. A constant prediction scores $1/K$, and for equal class sizes BA equals the accuracy.

The four-leaf tree reaches validation accuracy 0.8886 and BA 0.8686. Its depth was fixed in advance, so no selection took place, and it enters the final comparison unchanged with test accuracy 0.8861 and BA 0.8998. The BA gives the 59 cigar-shaped test galaxies the same weight as the 398 round ones.

Regression trees

The galaxy labels are categories. For a numerical target y, such as a disease-progression score, a tree divides the input space into leaf regions R_ℓ and assigns one constant c_ℓ to each region,

$$\widehat{f}(x) = \sum_{\ell} c_\ell \, \mathbf{1}\{x \in R_\ell\}.$$

Under squared-error loss, the best constant of a leaf is the mean of its fitting responses. Predictions are judged by the root mean squared error

$$\mathrm{RMSE} = \Big(\frac{1}{N} \sum_{i=1}^{N} (y_i - \widehat{y}_i)^2 \Big)^{1/2},$$

which carries the units of y. Many scientific targets are numbers, and a forest combines either numerical predictions or probability vectors.

Practice. Three numerical responses

Inputs $1, 2, 3$ have responses $1, 2, 6$. Compare thresholds 1.5 and 2.5.

1. Compute each child mean and total child squared error.
2. Choose a threshold and predict at input 2.2.

Feedback. The parent mean is 3. Child SSE sums are 8 and 0.5. The threshold 2.5 wins and predicts 1.5 at 2.2. The prediction is constant within a leaf.

Diabetes progression data

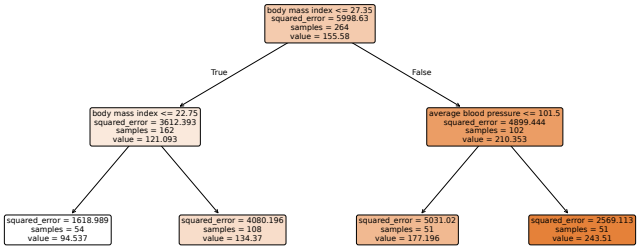
The diabetes data contain 442 patients with diabetes. For each patient ten variables were recorded at a baseline visit, and the target is a quantitative score of disease progression one year later. We use two inputs, body mass index (BMI, $\mathrm{kg/m^2}$) and average blood pressure (mm Hg).

BMI	Average blood pressure	Progression score
31.6	117	310
20.2	81	59
19.7	81	77
30.3	109	166

A random partition with seed 1975 gives 264 fitting, 89 validation and 89 test patients.

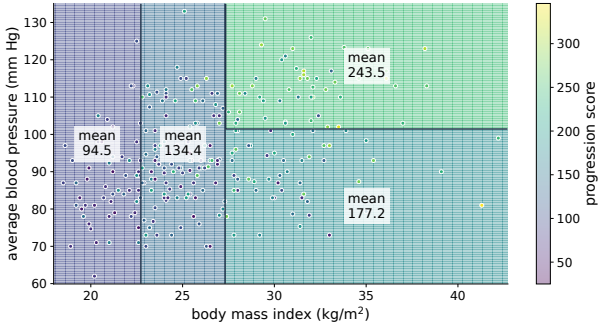
Scikit-learn diabetes data with unscaled values. B. Efron, T. Hastie, I. Johnstone, R. Tibshirani, Least angle regression, Ann. Stat. 32, 407 (2004)

The trained regression tree



Depth two, at least ten patients per leaf, 264 fitting patients. Leaf values are mean progression scores. Among heavier patients the tree asks about blood pressure, an interaction. The first validation patient, BMI 22.0, answers yes twice and receives 94.5, the mean of 54 fitting patients. The recorded score is 138.

The regression tree as boxes



Four leaves are four rectangles in the plane of BMI and blood pressure, coloured by the leaf means 94.5, 134.4, 177.2 and 243.5. The blood-pressure line at 101.5 mm Hg exists only for BMI above 27.35. The effect of blood pressure depends on BMI, an interaction that the tree represents by asking different questions in different branches.

After G. James, D. Witten, T. Hastie, R. Tibshirani, *An Introduction to Statistical Learning*, Springer (2013), Ch. 8, Fig. 8.2

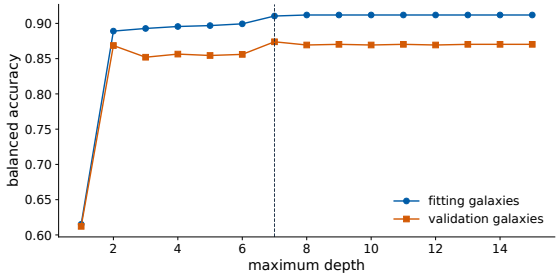
Eight descriptors per galaxy

The image measurement gives more numbers than q and C. Each galaxy becomes a row of eight columns.

Descriptor	Meaning
mask area fraction	fraction of the 4096 pixels inside the mask
brightness sum	summed background-subtracted display brightness in the mask
half-light radius r_{50}	radius enclosing half of that brightness, in pixels
axis ratio q	roundness from the brightness-weighted covariance
red-blue index	$(R - B)/(R + B)$ of the summed red (z-band) and blue (g-band) channels
concentration C	r_{90}/r_{50}
centre offset	distance of the brightness centre from the image centre, in pixels
mask failure	1 if no object was found at the centre

All are computed from the image without the label. The depth-two tree fitted to all eight columns is identical to the tree fitted to q and C.

Depth selection on validation galaxies



At least ten galaxies per leaf. The validation BA is 0.8686 at depth two (4 leaves) and reaches its largest value 0.8739 at depth seven (55 leaves). One of the 55 cigar-shaped validation galaxies moves the BA by 0.006. From depth eight the fitting BA stays at 0.9119, although the tree grows to 80 leaves at depth 13.

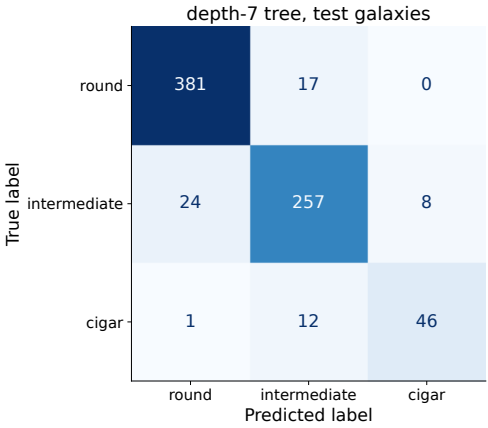
A shallow tree mixes classes in one leaf, a deep tree follows noise of the fitting sample. The fitting score keeps rising with depth, while the validation score barely moves after the second level. The extra 51 leaves describe details of the fitting galaxies.

Practice. Freeze a selection rule

Write down a choice of candidate depths, an evaluation score and a tie rule before examining test performance. Explain when a preference for a shorter explanation should enter the procedure.

Feedback. The declared maximum-validation rule selects depth seven. Depth two remains a separate teaching model fixed in advance. A simplicity preference belongs in the validation rule. Test results assess the frozen choice. A single rare-class object determines score granularity, which alone supplies no confidence interval.

The confusion matrix



Entry (k, l) counts the test galaxies of catalogue class k that the model assigns to class l. The diagonal holds the correct predictions, and BA is the mean of the diagonal divided by the row sums.

The depth-seven tree reaches test accuracy 0.9169 and BA 0.8754. The depth-two tree reaches 0.8861 and 0.8998. The deeper tree classifies more round galaxies correctly and fewer cigar-shaped ones.

Impurity importance

Each internal node t tests one feature j_t and contributes its impurity decrease $\Delta I(t)$, weighted by the fraction n_t/N of the N fitting observations that reach it,

$$A_j = \sum_{t:\ j_t = j} \frac{n_t}{N}\,\Delta I(t), \qquad \mathrm{Imp}_j = \frac{A_j}{\sum_{r=1}^{d} A_r}.$$

The sum over r runs over all d features. Classification trees use I_G and regression trees use the node MSE, for which $(n_t/N)\,\Delta I(t) = \Delta\mathrm{SSE}_t/N$.

Impurity importance is computed from the fitting data alone.

Permutation importance

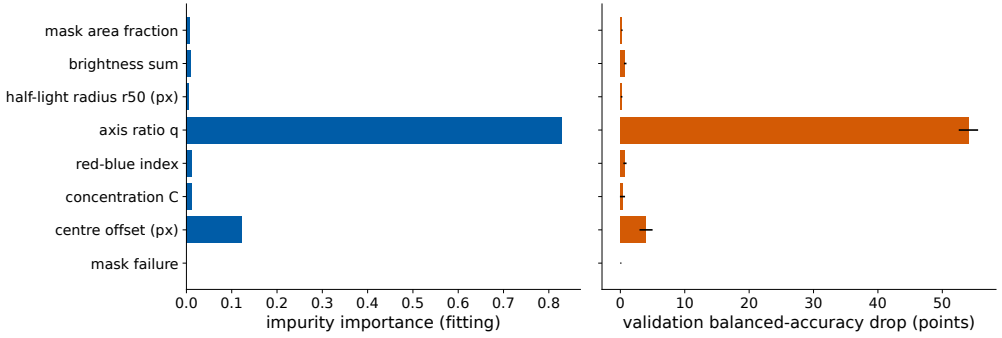
We shuffle feature j across the validation observations, keep the trained model fixed, and record the score $s_j^{(\kappa)}$ for shuffle $\kappa = 1, \ldots, n_{\mathrm{sh}}$. For a score s where larger is better,

$$\mathrm{PI}_j = s - \frac{1}{n_{\mathrm{sh}}} \sum_{\kappa=1}^{n_{\mathrm{sh}}} s_j^{(\kappa)},$$

with s the unshuffled validation score. For an error score such as the RMSE, the sign is reversed, so PI_j is the increase of the error. For the galaxies we use $n_{\mathrm{sh}} = 10$.

Shuffling keeps the distribution of feature j and destroys its relation to the target and to the other features. It can create input combinations that never occur in the data.

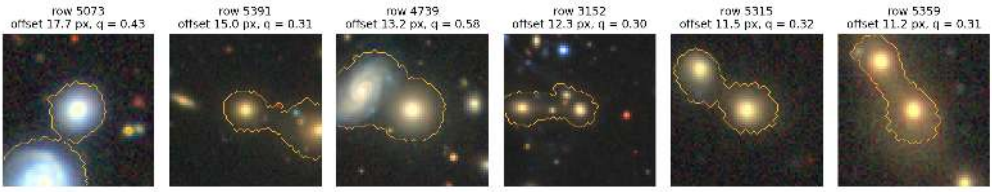
Feature importance in the galaxy tree



Depth-seven tree. Left, impurity importance from the fitting galaxies. Right, drop of the validation BA when one descriptor is shuffled, mean and standard deviation over ten shuffles. The axis ratio carries 0.83 of the impurity decrease and its shuffle lowers the validation BA by 54 points. The centre offset follows with 0.12 and 4 points.

The centre offset

The centre offset measures how far the brightness centre lies from the image centre. Offsets above 2 pixels occur for 9.6% of the round fitting galaxies, 6.5% of the intermediate and 5.0% of the cigar-shaped ones.



The six round fitting galaxies with the largest offsets. Each mask includes a neighbour, the centre moves between the two objects, and q falls to 0.30–0.58 for a round galaxy.

The offset reliance is consistent with marking unreliable measurements of q. The images support this explanation. A test would correct contaminated masks and compare the measurements, predictions and offset reliance on fixed evaluation objects.

A question that changes no prediction

The right-hand question $q \leq 0.823698$ of the depth-two galaxy tree sends the 1653 rounder galaxies into two leaves that both predict round, with round probabilities $76/109 = 0.70$ and $1518/1544 = 0.98$.

It lowers the Gini impurity of its node by 0.010093 and adds

$$\frac{1653}{3407} \cdot 0.010093 = 0.0049$$

to the impurity importance of q. No galaxy changes its predicted class, so the question adds nothing to accuracy, BA or any permutation importance computed from hard predictions. The log loss, which scores probabilities, registers it.

Random forests

A random forest combines predictions from many randomized trees. Scikit-learn handles their fitting internally.

- `RandomForestClassifier` predicts classes and probabilities.
- `RandomForestRegressor` predicts numerical responses.
- `n_estimators` sets the number of trees.

We use the same fitting, validation and test workflow as for one tree. A forest can be more stable, while shared measurement problems can still affect every tree.

Comparison on the same measurements

All models receive the same eight descriptors and the same data split. The dense network standardises its inputs using fitting statistics and learns class probabilities by minimizing cross-entropy. Its architecture and penalty are selected on validation observations. Detailed training equations are in the further material.

The comparison tests model families on this representation. A network learning its own features from image pixels is a separate experiment.

The same galaxies for three model families

We compare single trees, the forest and a dense neural network on the same eight descriptors and the same fitting, validation and test galaxies.

Each family chooses its settings on the validation galaxies. The imputation and the input scaling of the network use the fitting galaxies only.

The network receives the measured descriptors, as the trees do. A convolutional network that reads the pixels directly is a different model and was the subject of Lecture 4.

Galaxy measurements with three methods

The same eight measurements and the same test galaxies are supplied to the tree, forest and network.

Model	Test accuracy	Test BA
Tree	0.917	0.875
Forest	0.932	0.902
Network	0.948	0.937

The configurations are declared in the notebook. These reused course observations illustrate a comparison of fitted models.

A common application workflow

1. Inspect the inputs and target.
2. Separate fitting, validation and test observations.
3. Fit each scikit-learn estimator.
4. Compare validation scores and freeze the settings.
5. Evaluate test predictions and inspect individual errors.

A baseline gives the comparison scale. Keep preprocessing inside the network pipeline so it learns only from fitting observations.

MNIST images with three methods

Every model receives the same 784 scaled pixel values per image and the same fitting and test images.

Model	Test accuracy
Baseline	0.114
Tree	0.822
Forest	0.951
Network	0.942

The tree depth is selected on validation images. The forest and small dense network use declared settings. This is a small application with different tuning budgets.

Predictions on displayed digits



The same twelve test images. T, F and N denote tree, forest and network. Inspect a disagreement before interpreting the average accuracy.

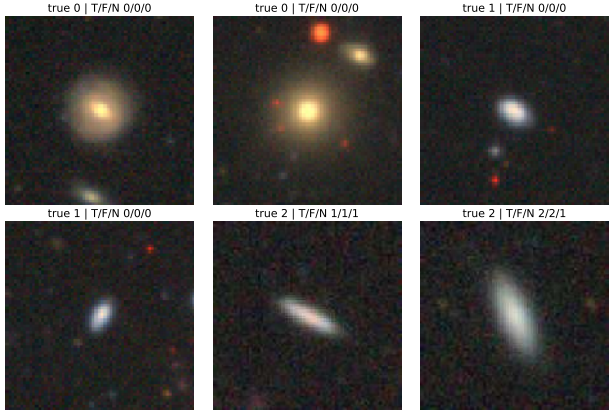
Galaxies classified directly from pixels

Every method receives the same 16×16 RGB block averages, flattened to 768 inputs. The target is round, intermediate or cigar-shaped.

Model	Test accuracy	Test BA
Tree	0.798	0.683
Forest	0.877	0.737
Network	0.879	0.678

These inputs differ from the eight measured descriptors. Image reduction keeps the example small and can discard fine structure.

Predictions on displayed galaxies



Two test images of each class, with tree, forest and network predictions. All methods receive the same reduced pixels. Codes 0, 1 and 2 mean round, intermediate and cigar-shaped.

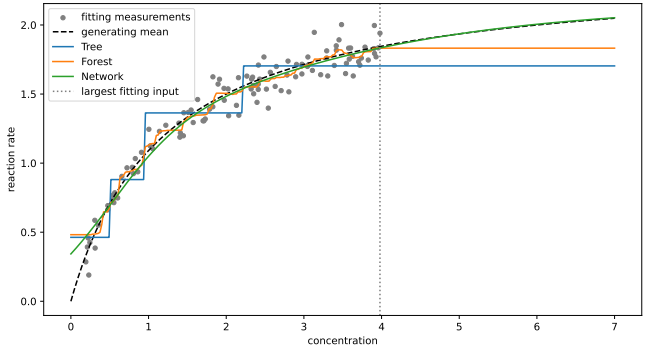
Numerical regression with three methods

Body mass index and blood pressure predict the same numerical progression score. Smaller RMSE means smaller error in score units.

Model	Test RMSE
Baseline	79.2
Tree	69.0
Forest	72.2
Network	67.5

Compare with the constant fitting-mean baseline. The same API workflow applies to classes and numbers.

A short concentration-response application



Fit a tree, forest and network with scikit-learn, then display their predictions. The dashed curve generated the measurements. Prediction outside the measured range needs separate evidence.

Explaining the learned model

Without consulting a formula, explain how a candidate question is scored and how a new observation receives its prediction.

- Distinguish class probabilities from the majority decision.
- Explain the roles of fitting, validation and test observations.
- Explain why many agreeing trees can still make the same mistake.
- Separate feature reliance, measurement artefacts and physical causes.

The practice feedback in the notes supplies numerical answers. Continue with the regression notebook and the forest comparison. The further material retains proofs and additional experiments.

Deriving the leaf mean

Leaf ℓ holds n_ℓ fitting responses with mean $\overline{y}_\ell$. Under squared loss a constant prediction c costs

$$S_\ell(c) = \sum_{i \in R_\ell} (y_i - c)^2, \qquad \frac{dS_\ell}{dc} = 2n_\ell(c - \overline{y}_\ell).$$

The derivative vanishes at $c = \overline{y}_\ell$, and $d^2S_\ell/dc^2 = 2n_\ell > 0$ makes this the unique minimum.

Absolute-error loss would give the median of the leaf.

Deriving the regression split criterion

Let $\mathrm{SSE}(A) = \sum_{i \in A} (y_i - \bar{y}_A)^2$ for a node A. A parent P with children L, R satisfies

$$\mathrm{SSE}(P) = \mathrm{SSE}(L) + \mathrm{SSE}(R) + n_L(\bar{y}_L - \bar{y}_P)^2 + n_R(\bar{y}_R - \bar{y}_P)^2,$$

because the deviations inside each child sum to zero. With $\bar{y}_P = (n_L \bar{y}_L + n_R \bar{y}_R)/n$ the decrease becomes

$$\Delta\mathrm{SSE} = \frac{n_L n_R}{n} (\bar{y}_L - \bar{y}_R)^2.$$

Dividing by n gives $\Delta\mathrm{SSE}/n$, the decrease of the node mean squared error, which is the local MSE decrease. The globally weighted decrease used by `min_impurity_decrease` is $\Delta\mathrm{SSE}/N$, with N the fitting-set size.

The class prediction of a leaf

Suppose leaf t predicts one class c for every observation that reaches it. Its fraction of fitting errors is

$$\frac{1}{n_t} \sum_{i \in t} \mathbf{1}\{y_i \neq c\} = 1 - p_{tc}.$$

The smallest error is obtained for the majority class $\widehat{y}_t = \arg\max_k p_{tk}$, and its value

$$I_M(t) = 1 - \max_k p_{tk}$$

is the misclassification error of the node. The leaf also stores the whole vector $(p_{t1}, \ldots, p_{tK})$ as estimated class probabilities.

Deriving the Gini decrease

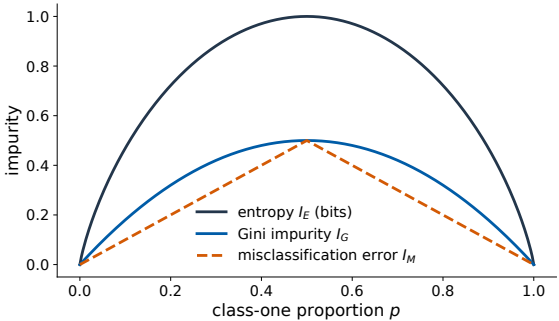
The parent proportions are the weighted mean $\boldsymbol{p}_t = \pi_L \boldsymbol{p}_L + \pi_R \boldsymbol{p}_R$. With $I_G = 1 - \|\boldsymbol{p}\|^2$ and $\pi_L + \pi_R = 1$,

$$\begin{aligned} \Delta I_G &= \pi_L \|\boldsymbol{p}_L\|^2 + \pi_R \|\boldsymbol{p}_R\|^2 - \|\pi_L \boldsymbol{p}_L + \pi_R \boldsymbol{p}_R\|^2 \\ &= \pi_L \pi_R \, \|\boldsymbol{p}_L - \boldsymbol{p}_R\|^2. \end{aligned}$$

Here $\|\boldsymbol{v}\|^2 = \sum_k v_k^2$. The second line uses $\pi_L - \pi_L^2 = \pi_L \pi_R$ and $\pi_R - \pi_R^2 = \pi_L \pi_R$.

The gain depends on both the difference between class proportions and the child sizes. Holding the difference fixed, equally sized children give the largest gain.

Gini impurity and misclassification error



For two classes with class-1 fraction p, the misclassification error and the Gini impurity are

$$I_M = \min(p, 1-p), \qquad I_G = 2p(1-p).$$

Splits that keep the majority class

The class-1 fractions of the children average to that of the parent, $p_t = \pi_L p_L + \pi_R p_R$.

I_M is piecewise linear. If both children keep the majority class of the parent, I_M is linear between p_L and p_R, and the split gives $\Delta I_M = 0$.

I_G is strictly concave, and for two classes

$$\Delta I_G = 2\pi_L \pi_R \left(p_L - p_R\right)^2 > 0 \quad \text{whenever } p_L \neq p_R.$$

Gini rewards a purer child before any prediction changes, and a later split can turn that purer child into correct predictions.

Shannon entropy

Averaging the information over the classes of node t gives the entropy

$$I_E(t) = -\sum_{k=1}^{K} p_{tk} \log_2 p_{tk}, \qquad 0 \le I_E(t) \le \log_2 K.$$

The proportion p_{tk} weights how often class k occurs, and $-\log_2 p_{tk}$ is its information.

A pure node has zero entropy, and equal proportions give the maximum $\log_2 K$.
Replacing I_G by I_E in the split objective gives the information gain.

Deriving the child weights

Select a fitting observation of node t uniformly at random. It goes left with probability n_L/n and right with probability n_R/n.

The expected impurity of the child it reaches is

$$\pi_L \, I(L) + \pi_R \, I(R) = \frac{n_L}{n} \, I(L) + \frac{n_R}{n} \, I(R).$$

The parent impurity $I(t)$ is the same for every candidate question. Minimising the expected child impurity is therefore equivalent to maximising ΔI.

Gini impurity as disagreement

Draw two labels independently from the same node, putting the first observation back before the second draw.

A pure node always gives two equal labels. A mixed node sometimes gives two different labels.

The Gini impurity is the probability that the two labels differ.

Thank You

Questions, project ideas, feedback: marcin.plodzien@uj.edu.pl

Website · GitHub · Google Scholar





Doctoral School of
Exact and Natural
Sciences